

RESEARCH PAPER COMPILATION

AI-Based Two-Wheeler Traffic Violation Detection, Helmet Detection, ALPR s Automated E-Challan Systems

Neatly organized research reference document
20 papers • 2022-2026

Prepared from the supplied research-paper dataset

1. Overview

Purpose. This document consolidates the supplied research papers into a clean, readable reference. For every paper, the title, takeaway, authors, year, citation count, abstract, study type, publication venue, SJR quartile (when supplied), DOI, and source link are retained.

2. Research Landscape at a Glance

Total papers: 20
Publication years represented: 2022, 2023, 2024, 2025, 2026
Most recent year represented: 2026
Common technical themes: YOLO-family object detection, OCR/EasyOCR/Tesseract, license plate recognition, DeepSORT/tracking, pose estimation, real-time video processing, and automated e-challan/notification workflows.

3. Comparative Research Matrix

No.	Year	Research Paper	Main Focus	Key Method / Result
1	2025	Automated traffic violation detection and E-challan system using YOLOv3 and IoT	This new traffic violation detection system using YOLOv3 and IoT effectively detects signal jumps and helmet violations in various environmental conditions, enabling intelligent traffic regulation in metropolitan areas.	YOLOv3, Tesseract, OCR, CRNN. This new traffic violation detection system using YOLOv3 and IoT effectively detects signal jumps and helmet violations in various environmental conditions, enabling intelligent tr
2	2026	Automated Road Safety Surveillance: Real-Time Helmet Detection and E-Challan Generation Using YOLO and OCR	This automated surveillance system detects helmetless riders and extracts vehicle registration numbers in real-time, helping authorities enforce helmet laws and promote safer riding behavior.	OCR. This automated surveillance system detects helmetless riders and extracts vehicle registration numbers in real-time, helping authorities enforce helmet laws and promote safer ridin
3	2025	A Deep Learning Approach for Helmet Detection and Fine Generation System	This study proposes a real-time deep learning-based helmet detection and fine generation system with over 95% accuracy, reducing manual workload and supporting road safety enforcement.	YOLOv3, OCR. accuracy of over 95% for helmet detection.
4	2023	Generating Smart E-challan System for Violation of Traffic Rules	This system automatically detects motorcyclists without helmets and retrieves their motorcycle number plates from CCTV footage, generating e-challans and databases for traffic rule violations.	OCR. This system automatically detects motorcyclists without helmets and retrieves their motorcycle number plates from CCTV footage, generating e-challans and databases for traffic rule
5	2025	Development of an AI-Based Smart Traffic Guard System for Real-Time Detection and Automated Penalty Enforcement of Traffic Violations		YOLOv5, OCR, Tkinter.
6	2025	Two-Wheeler Traffic Violations Detection and	This paper proposes an automated system that	This paper proposes an automated system that

		Automated Penalty Issuance System	detects traffic violations, tracks offenders, and issues tickets using vehicle registration numbers, improving enforcement efficiency and road safety.	detects traffic violations, tracks offenders, and issues tickets using vehicle registration numbers, improving enforcement efficiency a
7	2022	Two-Wheeler Vehicle Traffic Violations Detection and Automated Ticketing for Indian Road Scenario	This AI-based traffic violation detection system effectively detects two-wheeler violations and automates ticketing in Indian road scenarios, improving safety and promoting smart city ecosystems.	DeepSORT, Tesseract. precision (mAP) of 98.
8	2025	Computer-vision based automatic rider helmet violation detection and vehicle identification in Indian smart city scenarios using NVIDIA TAO toolkit and YOLOv8	Our two-step deep learning approach accurately detects rider helmet violations and vehicles in Indian smart cities, reducing traffic safety risks.	YOLOv8, OCR. achieved a promising helmet detection accuracy of 98.
9	2026	Modular AI-Based Framework for Two-Wheeler Traffic Violation Detection Using Object Detection, Pose Estimation, and ALPR	This modular AI-based system effectively detects two-wheeler traffic violations in real-time surveillance, generating e-challans, with a best mAP of 95%.	YOLOv8, YOLOv5, ALPR, pose estimation. achieved a best mAP of 95%, while the helmet detection model achieved 98.
10	2026	Real-Time YOLOv8-based Helmet Violation Detection and License Plate Recognition for Intelligent Traffic Monitoring	The proposed real-time traffic monitoring system accurately detects helmet violations and recognizes vehicle plates at a rate of 32 frames per second, providing safety in high-speed automated traffic monitoring.	YOLOv8. accuracy of 96.
11	2025	A Web-Based Ticketing System for Traffic Violations Using YOLO and OCR for Real-Time Detection	This web-based electronic ticketing system effectively detects traffic violations, particularly helmet noncompliance, with 93% helmet detection accuracy and 94% license plate detection accuracy.	OCR. accuracy, \$94.
12	2025	RoadEye: A Framework For Detecting Traffic Rule Violations	RoadEye, a framework for detecting traffic rule violations, shows promising accuracy in recognizing license plates, cross-walk violations, and helmetless driving.	mAP (Mean Accuracy Precision) score of 74% in detecting and localizing the license plates, 89% for cross-walk detection and 67% for helmetless driving detection.
13	2024	NUMBERPLATE DETECTION AND CHALLAN GENERATION FOR NON-HELMET RIDERS	This project develops a computerized CNN-based detection device for helmet authentication and a web-based interface for managing and tracking traffic infractions, improving safety and customer satisfaction.	This project develops a computerized CNN-based detection device for helmet authentication and a web-based interface for managing and tracking traffic infractions, improving safety
14	2026	SafeRide- AI Enabled Smart Road Safety Violation Detection and Monitoring System Using Neural Vision	This paper presents a real-time road safety violation detection system using deep learning and computer vision techniques, achieving high precision and recall,	YOLOv8. precision of 78%, recall of 79.

			potentially integrating with smart city infrastructure.	
15	2025	Automated Helmet Detection and Fine Issuance System Using YOLOv8 C OpenCV	The Helmet Detection for Motorcyclists using YOLOv8 and OpenCV system automatically identifies and penalizes violators in real-time, promoting helmet use and improving road safety.	YOLOv8, EasyOCR, OCR. The Helmet Detection for Motorcyclists using YOLOv8 and OpenCV system automatically identifies and penalizes violators in real-time, promoting helmet use and improving road safety.
16	2026	Smart Surveillance System for Triple Riding and No Helmet Violation Detection and Send Notification Using License Plate Recognition	This deep learning-based system effectively detects triple riding and helmetless riders using surveillance cameras, and automatically notifies registered vehicle owners of violations using license plate recognition.	YOLOv8, EasyOCR, OCR, CRNN. achieving robust detection under varying illumination and dense traffic conditions.
17	2025	Smart Traffic Surveillance: Helmet Detection and License Plate Recognition Using Deep Learning	This paper presents an advanced traffic violation detection system that combines helmet detection and license plate recognition using deep learning, enhancing road safety and reducing law enforcement burden.	YOLOv3, EasyOCR, OCR. accuracy by fine-hyperparameter tuning YOLOv3 on a diverse dataset that includes various helmet types, different lighting conditions, and occlusions.
18	2025	AI-Powered Smart Traffic Management System	AI-Powered Smart Traffic Management System effectively detects traffic violations and issues automated challans through Telegram, improving road safety and reducing human dependency in traffic monitoring.	accuracy and real-time performance, highlighting the potential for deployment in smart cities to improve road safety and reduce human dependency in traffic monitoring.
19	2023	Sighted-Helmet Detection and E-Challan Application	Helmet detection technology and e-challan systems can effectively enforce helmet laws and reduce head injuries in motorcycle accidents.	YOLOv8. Helmet detection technology and e-challan systems can effectively enforce helmet laws and reduce head injuries in motorcycle accidents.
20	2026	HELPR: Helmet Enforcement and License Plate Reconnaissance Using YOLOv8 and EasyOCR for Real-Time Traffic Surveillance	The HELPR system accurately detects helmet violations and motorcycle number plates in real-time traffic surveillance, offering scalable, data-driven traffic enforcement in smart cities.	YOLOv8, EasyOCR, OCR. accuracy in helmet detection, 96% accuracy in rider detection, and 88% accuracy in number plate detection, whereas EasyOCR gets a maximum of 96% accuracy in character-level OCR.

4. Detailed Research Papers

1. Automated traffic violation detection and E-challan system using YOLOv3 and IoT

Takeaway: This new traffic violation detection system using YOLOv3 and IoT effectively detects signal jumps and helmet violations in various environmental conditions, enabling intelligent traffic regulation in metropolitan areas.

Authors	G. Gambhire, Rohan Uplenchwar, Utkarsh Patil, Vansh Bhatt, Niraj Ukare, Apurva Waykande
Year	2025
Citations (as supplied)	1
Study Type	other
Journal / Conference	2025 IEEE International Conference on Internet of Things and Intelligence Systems (IoTaIS)
Journal SJR Quartile	Not supplied
DOI	10.1109/iotais67227.2025.11282128

Links: [DOI](#) | [Consensus paper page](#)

Abstract

This paper introduces a new version of real- time Traffic Rule Violation Detection and E-Challan Generation Application that is capable of detecting signal jumps and helmet violations under different environmental conditions at the same time. The proposed design leverages Multi-Scale Retinex with Color Restoration (MSRCR) and RGB-Thermal Image Fusion to bring out the features and increase the contrast of the images taken in low-light, rainy or foggy conditions. After that, the inputs are fed to the YOLOv3 Darknet-53 backbone, which helps in locating the riders without helmets and vehicles that are crossing the stop line at a red signal accurately. To identify the violators, the system uses a Hybrid OCR module consisting of Tesseract and CRNN with morphological filtering that can extract the characters of the number plates even in the case of blurred or damaged images. The system is an edge IoT-enabled setup, which uses a NodeMCU ESP8266 for operation, while the cloud-based connection with Firebase and its integration with flutter allow for the automated e-challan generation and real-time notification. The research findings demonstrate that there have been considerable improvements in the detection and the system’s real-time performance on low-power hardware has also been moderately enhanced. Such a system makes the way for intelligent traffic regulation in metropolitan areas.

2. Automated Road Safety Surveillance: Real-Time Helmet Detection and E-Challan Generation Using YOLO and OCR

Takeaway: This automated surveillance system detects helmetless riders and extracts vehicle registration numbers in real-time, helping authorities enforce helmet laws and promote safer riding behavior.

Authors	Pongiannan R K, P. S, V. B, P. Selvabharathi, P. M, R. L
Year	2026
Citations (as supplied)	0
Study Type	theoretical, modeling, or simulation study
Journal / Conference	2026 Third International Conference on Innovations in Cybersecurity and Data Science (ICICDS)
Journal SJR Quartile	Not supplied
DOI	10.1109/icicds70526.2026.11604897

Links: [DOI](#) | [Consensus paper page](#)

Abstract

This paper proposes an automated surveillance system for detecting helmetless riders and extracting vehicle registration numbers in real time. The system helps authorities to implement helmet laws because the manual human intervention is reduced and the motorcycles and riders can be followed in real-time video streams with the help of the methodology that is based on the object detection of the YOLO. When the system identifies a rider, then it immediately checks whether he or she is wearing a helmet or not. In case of a violation, the system focuses on the area of the license plate of the vehicle. The vehicle registration number is extracted using Optical Character Recognition (OCR) technology. All the corresponding information, such as photographs, timestamps, etc. are stored to be legally enforced, so that the administrator should be able to monitor the detections and examine the pattern of violation. The graphical interface was created on the basis of the PyQt to handle the data. It is a scalable and dependable design that may be smoothly incorporated into larger traffic surveillance systems. The proposed approach utilizes computer vision techniques to promote safer riding behavior and improve traffic surveillance efficiency.

3. A Deep Learning Approach for Helmet Detection and Fine Generation System

Takeaway: This study proposes a real-time deep learning-based helmet detection and fine generation system with over 95% accuracy, reducing manual workload and supporting road safety enforcement.

Authors	Sulbha Yadav, Sumit P. Singh, Dhruv Bedare, Ishan Samel
Year	2025
Citations (as supplied)	2
Study Type	theoretical, modeling, or simulation study
Journal / Conference	International Journal of Research and Innovation in Applied Science
Journal SJR Quartile	Not supplied
DOI	10.51584/ijrias.2025.10040076

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Traffic law violations, especially helmet non-compliance among motorcyclists, are a leading cause of road casualties in countries like India. This study proposes a real-time deep learning-based helmet detection and fine generation system that leverages YOLOv3 for object detection and OpenCV with OCR for license plate recognition. The system processes surveillance footage to detect violations and automatically generates e-challans by integrating with vehicle databases. Experimental results show an accuracy of over 95% for helmet detection. This automated solution reduces manual workload, supports road safety enforcement, and has the potential for integration into smart city infrastructure. Future work may involve multilingual plate recognition and improved database interoperability.

4. Generating Smart E-challan System for Violation of Traffic Rules

Takeaway: This system automatically detects motorcyclists without helmets and retrieves their motorcycle number plates from CCTV footage, generating e-challans and databases for traffic rule violations.

Authors	Harshada W. Nikam, Harshal Patil, Ashwini Ghodeswar, Siddhant Rajput, Prof. Amol Gosavi
Year	2023
Citations (as supplied)	2
Study Type	other
Journal / Conference	
Journal SJR Quartile	Not supplied

DOI	Not supplied
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Links: [Consensus paper page](#)

Abstract

- Motorcycles have always been the primary mode of transportation in developing countries. Motorcycle accidents have increased in recent years. One of the main reasons for fatalities in accidents is that a motorcyclist does not wear a protective helmet. The most common way to ensure that motorcyclists wear a helmet is by traffic police to manually monitor motorcyclists at road junctions or through CCTV footage and to penalize those without a helmet. But it requires human intervention and effort. So this system Proposes an automated system for detecting motorcyclists who do not wear a helmet and retrieving their motorcycle number plates from CCTV video footage. First, the system classifies moving objects as motorcycling or non-motorcycling. In the case of a classified motorcyclist, the head portion is located and classified as a helmet or non-helmet. Finally, for the motorcyclist identified without a helmet, the number plate of the motorcycle is detected and the characters on it are extracted by using the OCR algorithm. Echallan will also be generated with offender details. A database will be generated with records to identify every offender accurately. The system implements pure machine learning in order to identify every type of helmet that it comes across with minimum computation cost.

5. Development of an AI-Based Smart Traffic Guard System for Real-Time Detection and Automated Penalty Enforcement of Traffic Violations

Takeaway:

Authors	N. Thamilarasi, B. Shanmathi, S. Kannagi, G. M, S. M, S. P.
Year	2025
Citations (as supplied)	2
Study Type	
Journal / Conference	2025 6th International Conference on Inventive Research in Computing Applications (ICIRCA)
Journal SJR Quartile	Not supplied
DOI	10.1109/icirca65293.2025.11089648

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Presenting the design and Development of an AI-powered Traffic Guard system aimed at detecting and penalizing common traffic violations using an integrated tech stack. The system leverages YOLOv5 for real-time object detection, particularly helmet detection for two-wheeler riders. OpenCV, combined with OCR and License Plate Detection (ALPD) techniques, is utilized to extract vehicle registration numbers from live footage. Additionally, speed limit violations and red-light crossing incidents are detected using frame-based analysis and traffic signal synchronization. Once a breach is detected and verified, a digital challan is automatically generated and routed to the violator through Razorpay, enabling immediate online payment. This end-to-end solution is developed in Python, with a simple and intuitive Tkinter. GUI for monitoring and manual override, where necessary. The choice of tools ensures the system remains light, accurate, and easy to implement, even in resource-constrained environments. My focus is to maintain a balance of technical depth and realworld applicability while enhancing road safety and promoting regulatory compliance. Through this system, we aim to reduce human effort in traffic enforcement and foster a smarter and more disciplined urban transport environment.

6. Two-Wheeler Traffic Violations Detection and Automated Penalty Issuance System

Takeaway: This paper proposes an automated system that detects traffic violations, tracks offenders, and issues tickets using vehicle registration numbers, improving enforcement efficiency and road safety.

Authors	M. Pendyala, Rishitha Naik Pipavath
Year	2025
Citations (as supplied)	1
Study Type	other
Journal / Conference	International Journal of Research Publication and Reviews
Journal SJR Quartile	Not supplied
DOI	10.55248/gengpi.6.0525.18135

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Road safety is one of the highest priorities, as road accidents are among the major causes of deaths in India. Road Accidents are primarily caused due to violators of road safety regulations such as not wearing a helmet, triple riding etc. Though there are many smart systems to monitor these violations, yet they are prone to lot of errors. Additionally, the tickets are issued manually and this manual process mostly leads to delay and errors. In order to overcome these problems, we are suggesting an integrated system that automates the violation detection, makes tracking of offenders easier, and facilitates timely issuance and collection of fines. Such a system would increase enforcement efficiency, improve road safety, and minimize the frequency of accidents due to violation of traffic rules. The suggested system can identify whether the rider is wearing a helmet or not, identifies the pillion riders (not more than 2 persons including the rider), and even if they exceed the specified speed limit, under any violation of the above-stated rules the system can automatically issue tickets on the respective vehicles using its registration number (i.e., License plate). Though the ticket is issued, the majority of them won't pay the challan on time, therefore we introduce the concept of penalty points for each vehicle. If the penalty points cross a certain threshold, then we restrict the vehicle owner from further renewal of the vehicle insurance and even he/she will not be able to claim the insurance amount if needed.

7. Two-Wheeler Vehicle Traffic Violations Detection and Automated Ticketing for Indian Road Scenario

Takeaway: This AI-based traffic violation detection system effectively detects two-wheeler violations and automates ticketing in Indian road scenarios, improving safety and promoting smart city ecosystems.

Authors	R. Charran, R. Dubey
Year	2022
Citations (as supplied)	58
Study Type	theoretical, modeling, or simulation study
Journal / Conference	IEEE Transactions on Intelligent Transportation Systems
Journal SJR Quartile	1.0
DOI	10.1109/tits.2022.3186679

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Traffic violation monitoring and control is a major concern in India due to excess crowd, increasing commuters, bad traffic signal management, and rider mentality. It is obvious that physical traffic police-based monitoring alone is insufficient to monitor such large traffic volumes and simultaneously track violations. This has led to many violators going unnoticed. The violators, in turn, cause more serious mishaps on the road resulting in

danger to their own life as well as to other’s life. Thus, there is a need for incorporating Artificial Intelligence (AI)-based techniques to eliminate manual intervention for the detection and catching of violators. In this paper, we propose a system to automatically detect two-wheeler violations like not wearing a helmet, usage of a phone while riding, triple riding, wheeling, and illegal parking for Indian road scenarios and eventually automating the ticketing process by capturing the violations and corresponding vehicle number in a database. We propose using a custom trained Yolo-v4 + DeepSORT for violation detection and tracking and Yolo-v4 + Tesseract for number plate detection and extraction. This implementation obtained a mean average precision (mAP) of 98.09% for violation detection and an accuracy of 99.41% for number plate detection on the test data. Further, the system detected 77 out of 93 violations with zero false positives in real-life scenarios. Thus, showing that the traffic violation system developed can be used to automate traffic violation ticketing. The developed system would be particularly useful in deriving various safety-related policies and will help to enforce strong regulation of traffic rules and build towards a smart city ecosystem via the automated AI-based traffic violation and ticketing system.

8. Computer-vision based automatic rider helmet violation detection and vehicle identification in Indian smart city scenarios using NVIDIA TAO toolkit and YOLOv8

Takeaway: Our two-step deep learning approach accurately detects rider helmet violations and vehicles in Indian smart cities, reducing traffic safety risks.

Authors	Uttam U. Deshpande, Goh Kah Ong Michael, Sufola Das Chagas Silva Araujo, Vaidehi P Deshpande, Rudragoud Patil, Ramchandra Alias Ameet Chate, Varun R. Tandur, Supreet S. Goudar, Shreya Ingale, Vaishnavi Charantimath
Year	2025
Citations (as supplied)	8
Study Type	theoretical, modeling, or simulation study
Journal / Conference	Frontiers in Artificial Intelligence
Journal SJR Quartile	2.0
DOI	10.3389/frai.2025.1582257

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Two-wheeler traffic offenses are a well-known fact about the Indian Road scenario. In addition to endangering the offenders, these offenses also endanger other commuters. Two-wheeler traffic violations can take many different forms, such as overloading, triple riding, and helmetless riding. Effective identification and enforcement strategies are necessary for these offenses since they pose a serious risk to public safety. Due to the inadequacy of traditional traffic monitoring and enforcement techniques, advanced technology-based solutions are now required. Deep learning-based systems have demonstrated significant promise in identifying and stopping such infractions in recent years. We propose a two-step deep learning approach that leverages the strengths of pre-trained object detection models to detect two-wheeler riders and specialized helmet classifiers to identify helmet wear status as well as detect number plates. In the first stage, we utilized a highly efficient, robust, and accurate object identification DetectNet (Model 1) framework developed by NVIDIA, and it uses the ResNet18 Convolutional Neural Network (CNN) architecture as part of the Transfer Learning Toolkit known as TAO (Train, Adapt, Optimize). The second stage demands accurate detection of a helmet on the identified rider and extracting numbers from the violator’s license plates using the OCR module in real time. We employed YOLOv8 (Model 2), a deep learning-based architecture that has proven effective in several applications involving object detection in real time. It predicts bounding boxes and class probabilities for objects within an image using a single neural network, making it a perfect choice for real-time applications like rider helmet violations detections and number plate processing. Due to a lack of publicly available traffic datasets, we created a custom dataset containing

motorcycle rider images captured under complex scenarios for training and validating our models. Experimental analysis shows that our proposed two-step model achieved a promising helmet detection accuracy of 98.56% and a 97.6% number plate detection accuracy of persons not wearing helmets. The major objective of our proposed study is to enforce stringent traffic laws in real-time to decrease rider helmet violations.

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9. Modular AI-Based Framework for Two-Wheeler Traffic Violation Detection Using Object Detection, Pose Estimation, and ALPR

Takeaway: This modular AI-based system effectively detects two-wheeler traffic violations in real-time surveillance, generating e-challans, with a best mAP of 95%.

Authors	Bondita Paul, Sangamlung Gangmei, Simanta Hazarika, Vikosalie Chusi, Deep Das, Marcel Basumatary, Chiranjeeta Borah
Year	2026
Citations (as supplied)	0
Study Type	theoretical, modeling, or simulation study
Journal / Conference	2026 International Conference on Cognitive Computing and Networking Systems (ICC-CNS)
Journal SJR Quartile	Not supplied
DOI	10.1109/icc-cns70518.2026.11606049

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Rapid growth in urban traffic has made the manual monitoring of traffic-rule violations increasingly difficult, particularly for two-wheeler offenses such as helmetless riding and triple riding. This paper presents a modular AI-based automated enforcement system for detecting two-wheeler traffic violations from real-time surveillance and generating challans automatically. The proposed system processes live video streams through sequential pre-processing, two-wheeler detection, helmet violation detection, rider counting through pose estimation, and automatic license plate recognition (ALPR), followed by violation aggregation and e-challan generation. To support this framework, multiple deep learning models were trained and evaluated for core subtasks. The developed two-wheeler detection model using YOLOv8s achieved a best mAP of 95%, while the helmet detection model achieved 98.2% mAP. For license plate extraction, a YOLOv8n-based ALPR model outperformed a YOLOv5 baseline on the Indian license plate dataset, achieving 99.3% precision, 99.8% recall, and 99.5% [mAP@0.5](#), while also reducing model size and improving inference efficiency. In addition, a pretrained YOLOv8 pose model is incorporated for rider counting to identify triple riding cases. The entire framework is designed to be cost-effective, scalable, and suitable to integrate with real-time dashboards and automated traffic enforcement workflows. The proposed system demonstrates the possibility of combining object detection, pose estimation, and license plate recognition into a unified intelligent traffic monitoring solution.

10. Real-Time YOLOv8-based Helmet Violation Detection and License Plate Recognition for Intelligent Traffic Monitoring

Takeaway: The proposed real-time traffic monitoring system accurately detects helmet violations and recognizes vehicle plates at a rate of 32 frames per second, providing safety in high-speed automated traffic monitoring.

Authors	G. M. Babu, Komera Anantu Babu, Reddy Vari Teja Reddy, Palamuri Bhanu Kumar Reddy
Year	2026
Citations (as supplied)	0
Study Type	theoretical, modeling, or simulation study
Journal / Conference	2026 7th International Conference on Inventive Research in Computing Applications (ICIRCA)
Journal SJR Quartile	Not supplied
DOI	10.1109/icirca69024.2026.11570524

Links: [DOI](#) | [Consensus paper page](#)

Abstract

In Traditional traffic surveillance depends on human observation that can usually have inconsistent enforcement and cannot cover extensive areas of activity in case of a surge in traffic. In an attempt to solve these problems, this paper will propose a real time smart traffic monitoring system that will be capable of identifying helmet offending and automatically recognize the vehicle grant numbers. The proposed system involves the YOLOv8 deep learning detection model and has a multi-stage processing approach. The main steps that are followed are as follows: first, traffic video frames have to be analyzed regarding whether a rider wears a helmet or not. In the event of breach, the system identifies the plate of vehicles plate, the registration number is identified and major violation is automatically registered. he researchers used a manually annotated dataset of 8,460 images for testing and training the system using realistic road conditions. The dataset possesses variations in terms of various lightings, locations of the riders, partial occlusions, and motion blur. The experimental results also suggest that the proposed method can deliver the detection accuracy of 96.8 percent and process the rate at about 32 frames per second, which can be considered an effective solution to the problem of providing the safety in a high-speed automated traffic monitoring system.

11. A Web-Based Ticketing System for Traffic Violations Using YOLO and OCR for Real-Time Detection

Takeaway: This web-based electronic ticketing system effectively detects traffic violations, particularly helmet noncompliance, with 93% helmet detection accuracy and 94% license plate detection accuracy.

Authors	Avicena Naufaldo, H. Fakhurroja, Dita Pramesti
Year	2025
Citations (as supplied)	0
Study Type	other
Journal / Conference	2025 5th International Conference on Electronic and Electrical Engineering and Intelligent System (ICE3IS)
Journal SJR Quartile	Not supplied
DOI	10.1109/ice3is66769.2025.11281307

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Traffic violations, particularly helmet noncompliance, are a leading cause of road accidents. This study presents a web-based electronic ticketing system that enhances enforcement through violation detection. The system utilizes Raspberry Pi 5 with Pi Camera and an embedded YOLOv11 model for helmet detection, while OCR extracts license plate numbers. Relevant data is transmitted via API to a backend server, which handles notifications (SMS, WhatsApp, email) and fine payments through a gateway. Validation testing showed 93% helmet detection accuracy, 94.36% license plate detection accuracy, and a 9% OCR error rate. Usability

testing with 20 vehicle owners and 3 police officers yielded SUS scores of 72.88 and 85, along with positive SEQ ratings, demonstrating the system’s effectiveness in supporting enforcement and user workflows.

12. RoadEye: A Framework For Detecting Traffic Rule Violations

Takeaway: RoadEye, a framework for detecting traffic rule violations, shows promising accuracy in recognizing license plates, cross-walk violations, and helmetless driving.

Authors	A. Dixit, Ravi Bhandari, Pallab Maji
Year	2025
Citations (as supplied)	2
Study Type	
Journal / Conference	Proceedings of the 26th International Conference on Distributed Computing and Networking
Journal SJR Quartile	Not supplied
DOI	10.1145/3700838.3703683

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Traffic rule violations by vehicles is a serious issue and a cause of many road accidents worldwide. This paper presents preliminary results towards development of a framework, called RoadEye, for detecting traffic violations and subsequently generate e-challans. We have developed novel cross-walk and helmetless driving detectors, and used them to demonstrate the RoadEye pipeline, all the way from detection of violating vehicles to reading the license plate of the vehicle. Our evaluations backed by extensive data collection on chaotic and busy roads, obtained an accuracy of 82% in recognizing the license plates, mAP (Mean Accuracy Precision) score of 74% in detecting and localizing the license plates, 89% for cross-walk detection and 67% for helmetless driving detection.

13. NUMBERPLATE DETECTION AND CHALLAN GENERATION FOR NON-HELMET RIDERS

Takeaway: This project develops a computerized CNN-based detection device for helmet authentication and a web-based interface for managing and tracking traffic infractions, improving safety and customer satisfaction.

Authors	
Year	2024
Citations (as supplied)	0
Study Type	other
Journal / Conference	International Research Journal of Modernization in Engineering Technology and Science
Journal SJR Quartile	Not supplied
DOI	10.56726/irjmet52280

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Over time, there has been an increase in motorbike accidents in a number of nations. The automated detection of traffic rule violators is a prerequisite for any intelligent traffic system. With a high population density in all of the major cities, motorcycles are one of the primary modes of transportation in a country like India. In India, there are more than 37 million two-wheeler riders. According to statistics, the majority of motorbike riders don't wear helmets in cities or even on highways. Wearing a helmet can reduce a rider's risk of incurring a head injury or catastrophic brain injury in the majority of motorcycle accident scenarios. For the sake of traffic safety, a system

for automatically identifying helmets is therefore necessary. As a result, customized object detection models are constructed with a CNN-based algorithm (YOLO). When a helmetless rider is detected, the number plate is collected and the license registration number is identified using an optical character recognition system. The objective of this project is to design a computerized CNN-based detection device with custom-trained models and datasets for helmet authentication. Additionally, the Challan System is a web-based interface intended to assist users with various matters pertaining to managing and tracking traffic infractions. Essentially, the Challan System makes it possible for bike riders and police to have straightforward conversations using a web-based interface. This project concept demonstrates how users can complete tasks more quickly when Challan is available online. The web-based approach aims to minimize paperwork and conventional procedures in order to improve customer satisfaction.

14. SafeRide- AI Enabled Smart Road Safety Violation Detection and Monitoring System Using Neural Vision

Takeaway: This paper presents a real-time road safety violation detection system using deep learning and computer vision techniques, achieving high precision and recall, potentially integrating with smart city infrastructure.

Authors	Veena Trivedi, Priyanka Barman, Swati Parhad, Sakshi Kadam, Radhika Pradhan, Tejas Deshmukh
Year	2026
Citations (as supplied)	0
Study Type	theoretical, modeling, or simulation study
Journal / Conference	2026 2nd International Conference on Computing, Communication and Green Engineering (CCGE)
Journal SJR Quartile	Not supplied
DOI	10.1109/ccge67142.2026.11581600

Links: [DOI](#) | [Consensus paper page](#)

Abstract

The enforcement of traffic rules is often inconsistent and highly dependent on manual monitoring, resulting in delayed response and thus a high risk of accidents. This paper proposes a system for real-time road safety violation detection using deep learning and computer vision techniques. In the proposed system, the framework is based on the use of YOLOv8 for multi-class object detection. Automatic Number Plate Recognition is integrated for the identification of the vehicles. The system is capable of detecting various traffic violations, including helmetless driving, triple riding, seatbelt violations, red light violations, wrong-side driving, and mobile phone use while driving along with processing live video feeds in real-time and automatically logging the violations with timestamp and type of violation along with the analytical dashboard, thus enabling automated e-challans. It supports live streaming and manual video uploading. A high precision of 78%, recall of 79.1%, [mAP@0.5](#) of 82.3%, and [mAP@0.5-0.95](#) of 52.7% was obtained. This system has the potential for integration with smart city infrastructure.

15. Automated Helmet Detection and Fine Issuance System Using YOLOv8 & OpenCV

Takeaway: The Helmet Detection for Motorcyclists using YOLOv8 and OpenCV system automatically identifies and penalizes violators in real-time, promoting helmet use and improving road safety.

Authors	Hirald Dwaraka Praveena, Goruva Vardhan Babu, Jinka Muni Sai Yateesh, Degasikshi Guna, G. Krishna
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Year	2025
Citations (as supplied)	4
Study Type	theoretical, modeling, or simulation study
Journal / Conference	2025 3rd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)
Journal SJR Quartile	Not supplied
DOI	10.1109/idciot64235.2025.10915081

Links: [DOI](#) | [Consensus paper page](#)

Abstract

In densely populated countries like India, traffic violations are a major concern, especially among motorcyclists, leading to accidents and significant loss of life and property. A prevalent issue is the failure to wear helmets. To address this, a system called “Helmet Detection for Motorcyclists using YOLOv8 and OpenCV” is proposed, leveraging deep learning to automatically identify and penalize violators in real-time through traffic cameras. The system employs the You Only Look Once (YOLO) algorithm, a deep neural network-based object detection model, to identify riders without helmets. Subsequently, it detects the number plate of violators, extracting its text using the EasyOCR(Optical Character Recognition) library. This extracted data is cross-referenced with a pre-existing database to retrieve the owner's contact details. For automation, an SMS API(Application Programming Interface) is integrated to send fine notifications to the violator's registered mobile number. By encouraging helmet use, the system aims to minimize head injuries during accidents and improve road safety. This automated approach enhances traffic law enforcement, provides evidence of violations, and ensures efficient penalty issuance. Ultimately, the system supports safer roads by promoting compliance with traffic regulations and protecting motorcyclists' lives.

16. Smart Surveillance System for Triple Riding and No Helmet Violation Detection and Send Notification Using License Plate Recognition

Takeaway: This deep learning-based system effectively detects triple riding and helmetless riders using surveillance cameras, and automatically notifies registered vehicle owners of violations using license plate recognition.

Authors	Mohazzeba Tanveer Raza, Shilpa, S. N., Sridevi, Swapna Ujjappa Arikatti
Year	2026
Citations (as supplied)	0
Study Type	theoretical, modeling, or simulation study
Journal / Conference	2026 3rd International Conference on Emerging Trends in Engineering and Medical Sciences (ICETEMS)
Journal SJR Quartile	Not supplied
DOI	10.1109/icetems66917.2026.11469557

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Traffic violations such as triple riding and riding without helmets are major contributors to road accidents and fatalities. Manual monitoring of these violations is inefficient and error prone, highlighting the need for automated surveillance solutions. This paper presents a deep learning-based system that leverages surveillance cameras to detect triple riding and helmetless riders, integrated with license plate recognition for offender identification. In the proposed system, YOLOv8 is utilized for real-time identification of motorcyclists and helmet compliance, whereas licence plate recognition is carried out through EasyOCR integrated with a Convolutional Recurrent Neural Network (CRNN) model. A prototype database is incorporated to simulate the automated process of

sending violation alerts and fines to registered vehicle owners. Experimental evaluations on traffic surveillance footage demonstrate the system's effectiveness, achieving robust detection under varying illumination and dense traffic conditions. The proposed framework contributes to the development of Intelligent Transportation Systems (ITS), providing a scalable and practical solution to assisting law enforcement agencies in enhancing road safety through automated violation monitoring.

17. Smart Traffic Surveillance: Helmet Detection and License Plate Recognition Using Deep Learning

Takeaway: This paper presents an advanced traffic violation detection system that combines helmet detection and license plate recognition using deep learning, enhancing road safety and reducing law enforcement burden.

Authors	Patlolla Sathvika Reddy
Year	2025
Citations (as supplied)	0
Study Type	theoretical, modeling, or simulation study
Journal / Conference	INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT
Journal SJR Quartile	Not supplied
DOI	10.55041/ijjsrem46068

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Abstract— Motorcyclists are particularly vulnerable to road accidents, often resulting in severe injuries or fatalities. Helmets are a proven safety measure that significantly reduces the risk of head injuries; however, many riders fail to comply with helmet laws, making enforcement a challenge for traffic authorities. This paper presents an advanced, automated traffic violation detection system that integrates real-time helmet detection and automatic license plate recognition to enhance road safety and streamline enforcement. Our system employs an optimized YOLOv3 model to efficiently detect motorcycles and determine whether riders are wearing helmets. Unlike conventional implementations, we enhance detection accuracy by fine-hyperparameter tuning YOLOv3 on a diverse dataset that includes various helmet types, different lighting conditions, and occlusions. To address challenges such as low-light environments, we incorporate preprocessing techniques, including contrast enhancement and adaptive thresholding, improving detection performance under suboptimal conditions. For license plate recognition, we utilize EasyOCR, further improved through custom preprocessing steps such as noise reduction and edge enhancement and thresholding enabling better recognition of partially occluded or low-quality license plates. Upon detecting a violation, the system automatically extracts and logs the license plate details into a structured database, facilitating streamlined enforcement and legal action. Our experimental results demonstrate increased accuracy while reducing inference time compared to existing methods, making this system a scalable and deployable solution for real-time traffic monitoring. By automating violation detection and reporting, the proposed approach reduces the burden on law enforcement while encouraging greater compliance with helmet laws, ultimately contributing to safer roads.

Keywords— YOLO, EasyOCR, preprocessing techniques, traffic regulations and real-time monitoring.

18. AI-Powered Smart Traffic Management System

Takeaway: AI-Powered Smart Traffic Management System effectively detects traffic violations and issues automated challans through Telegram, improving road safety and reducing human dependency in traffic monitoring.

Authors	B. C
Year	2025
Citations (as supplied)	2
Study Type	theoretical, modeling, or simulation study
Journal / Conference	INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT
Journal SJR Quartile	Not supplied
DOI	10.55041/ijrsrem47454

Links: [DOI](#) | [Consensus paper page](#)

Abstract

ABSTRACT Urban traffic congestion and rule violations remain significant challenges for modern cities. This paper proposes an AI-Powered Smart Traffic Management System that integrates real-time video analysis, intelligent violation detection, license plate recognition, and automated challan issuance through Telegram. The system uses computer vision and machine learning algorithms to identify traffic violations such as over-speeding, helmet non-compliance, seatbelt negligence, and red-light jumping. License Plate Recognition (LPR) is used to identify the vehicle, and an automated challan is sent to the violator. A Telegram bot is integrated for real-time notification, and a back-end database maintains logs. Experimental results demonstrate high accuracy and real-time performance, highlighting the potential for deployment in smart cities to improve road safety and reduce human dependency in traffic monitoring. **Keywords** - Deep Learning, Natural Language Processing, Symptomatic Neural Network, machine learning, Mental Health Chat bot, Text. and Speech Processing, Depression Detection

Keywords—Smart traffic, AI, deep learning, license plate recognition, traffic violation, e-challan, Telegram bot, computer vision.

19. Sighted-Helmet Detection and E-Challan Application

Takeaway: Helmet detection technology and e-challan systems can effectively enforce helmet laws and reduce head injuries in motorcycle accidents.

Authors	Ritesh Mahajan, A. Patil, S. Sandhu, O. Telang, Niranjan S. Samudre
Year	2023
Citations (as supplied)	1
Study Type	other
Journal / Conference	International Journal for Research in Applied Science and Engineering Technology
Journal SJR Quartile	Not supplied
DOI	10.22214/ijraset.2023.50858

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Abstract: Helmet detection is a technology that uses computer vision algorithms to automatically detect and identify individuals who are not wearing helmets while riding a two-wheeled vehicle. This technology is often used in conjunction with e-challan systems, which are electronic systems for issuing traffic violations and fines. By combining helmet detection with e-challan, law enforcement officials can more effectively enforce helmet laws

and reduce the number of injuries and deaths caused by head injuries in motorcycle accidents. The technology can be used in various forms like CCTV cameras, drones, etc. The Helmet detection algorithm uses deep learning techniques like CNN, YOLOv8 etc to detect and classify the helmets in real-time video feed. We have developed an application “Sighted”, which is used to track the current active challans issued in the name of the owner and easy payment of the fines. The main objective of the system is to identify riders without helmet and issue a challan with an application providing services to track the challan status and payment.

20. HELPR: Helmet Enforcement and License Plate Reconnaissance Using YOLOv8 and EasyOCR for Real-Time Traffic Surveillance

Takeaway: The HELPR system accurately detects helmet violations and motorcycle number plates in real-time traffic surveillance, offering scalable, data-driven traffic enforcement in smart cities.

Authors	Kumar Manthri, S. Sai Samay, T. N. Chandra Sahith, A. T
Year	2026
Citations (as supplied)	2
Study Type	theoretical, modeling, or simulation study
Journal / Conference	2026 7th International Conference on Mobile Computing and Sustainable Informatics (ICMCSI)
Journal SJR Quartile	Not supplied
DOI	10.1109/icmcsi67283.2026.11412521

Links: [DOI](#) | [Consensus paper page](#)

Abstract

Traffic accidents that result in death of people are mainly caused by motorcycle accidents, especially in those areas where wearing of helmets is not a mandatory practice. To solve the problem of manual surveillance by the traffic authorities, this paper suggests an automated, real-time possession to identify the helmet violation and motorcycle number plates on the roadside surveillance streams. The framework incorporates the YOLOv8 object detector model to localize riders, helmets, and number plates, and uses the EasyOCR to do strong optical character recognition on plate regions extracted under a variety of lighting and traffic conditions. Experimental results on controlled datasets and on real-world video footage show that the system can be characterized by a high detection rate, achieving 93% accuracy in helmet detection, 96% accuracy in rider detection, and 88% accuracy in number plate detection, whereas EasyOCR gets a maximum of 96% accuracy in character-level OCR. a highquality text extraction rate, and a high generalization ability to a variety of complex traffic scenes, which means that the system has an opportunity to provide scalable, data-driven traffic enforcement within the smart city environment.

5. Quick Reference: Paper Links

1. Automated traffic violation detection and E-challan system using YOLOv3 and IoT – [DOI](#) | [Consensus](#)
2. Automated Road Safety Surveillance: Real-Time Helmet Detection and E-Challan Generation Using YOLO and OCR – [DOI](#) | [Consensus](#)
3. A Deep Learning Approach for Helmet Detection and Fine Generation System – [DOI](#) | [Consensus](#)
4. Generating Smart E-challan System for Violation of Traffic Rules – [Consensus](#)
5. Development of an AI-Based Smart Traffic Guard System for Real-Time Detection and Automated Penalty Enforcement of Traffic Violations – [DOI](#) | [Consensus](#)
6. Two-Wheeler Traffic Violations Detection and Automated Penalty Issuance System – [DOI](#) | [Consensus](#)
7. Two-Wheeler Vehicle Traffic Violations Detection and Automated Ticketing for Indian Road Scenario – [DOI](#) | [Consensus](#)
8. Computer-vision based automatic rider helmet violation detection and vehicle identification in Indian smart city scenarios using NVIDIA TAO toolkit and YOLOv8 – [DOI](#) | [Consensus](#)
9. Modular AI-Based Framework for Two-Wheeler Traffic Violation Detection Using Object Detection, Pose Estimation, and ALPR – [DOI](#) | [Consensus](#)
10. Real-Time YOLOv8-based Helmet Violation Detection and License Plate Recognition for Intelligent Traffic Monitoring – [DOI](#) | [Consensus](#)
11. A Web-Based Ticketing System for Traffic Violations Using YOLO and OCR for Real-Time Detection – [DOI](#) | [Consensus](#)
12. RoadEye: A Framework For Detecting Traffic Rule Violations – [DOI](#) | [Consensus](#)
13. NUMBERPLATE DETECTION AND CHALLAN GENERATION FOR NON-HELMET RIDERS – [DOI](#) | [Consensus](#)
14. SafeRide- AI Enabled Smart Road Safety Violation Detection and Monitoring System Using Neural Vision – [DOI](#) | [Consensus](#)
15. Automated Helmet Detection and Fine Issuance System Using YOLOv8 s OpenCV – [DOI](#) | [Consensus](#)
16. Smart Surveillance System for Triple Riding and No Helmet Violation Detection and Send Notification Using License Plate Recognition – [DOI](#) | [Consensus](#)
17. Smart Traffic Surveillance: Helmet Detection and License Plate Recognition Using Deep Learning – [DOI](#) | [Consensus](#)
18. AI-Powered Smart Traffic Management System – [DOI](#) | [Consensus](#)
19. Sighted-Helmet Detection and E-Challan Application – [DOI](#) | [Consensus](#)
20. HELPR: Helmet Enforcement and License Plate Reconnaissance Using YOLOv8 and EasyOCR for Real-Time Traffic Surveillance – [DOI](#) | [Consensus](#)

6. Notes

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